# Predicting Core Level Binding Energies Shifts: Suitability of the Projector Augmented Wave Approach as Implemented in VASP

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Here, we assess the accuracy of various approaches implemented in Vienna *ab initio* simulation package code to estimate core-level binding energy shifts (ΔBEs) using a projector augmented wave method to treat core electrons. The performance of the Perdew–Burke–Ernzerhof (PBE) and the Tao–Perdew–Staroverov–Scuseria (TPSS) exchange-correlation density functionals is examined on a dataset of 68 molecules containing B→F atoms in diverse chemical environments, accounting for 185 different 1*s* core level binding energy shifts, for which both experimental gas-phase X-ray photoemission (XPS) data and accurate all electron ΔBEs are available. Four procedures to calculate core-level shifts are investigated. Janak–Slater transition state approach yields mean absolute errors of 0.37 (0.21) eV at PBE (TPSS) level, similar to highly accurate all electron ΔSCF approaches using same functionals, and close to XPS experimental accuracy of 0.1 eV. The study supports the use of these procedures to assign ΔBEs of molecular moieties on material surfaces of interest in surface science, nanotechnology, and heterogeneous catalysis. © 2017 Wiley Periodicals, Inc.

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## Introduction

X-ray photoelectrons spectroscopy (XPS)—a.k.a. electron spectroscopy for chemical analysis—is nowadays a widespread technique used for material bulk elemental analysis, especially in surface science studies, given its remarkable surface sensitivity.[1] Indeed, it is not only sensitive to a particular material surface termination, but to the atoms and molecules adsorbed onto, which makes it a well-suited technique to detect surface species,[2] and even capable to discern adsorption conformations.[3,4] The detection of emerged electrons over time allows to *in situ* follow a given surface process, from adsorption and diffusion, to surface reactions heterogeneously catalyzed by the substrate, enabling the characterization of reactants, intermediates, and products, and, ultimately, getting information of the reaction mechanism and kinetics.[2,5,6]

The discrimination of one from another species (or conformations) links to particular atomic core level (CL) binding energies and, more importantly, CL binding energy shifts with respect to a given reference, which provide information about different chemical environments and/or oxidation states. The development of high resolution XPS (HR-XPS) technique allows nowadays to distinguish core binding energy shifts (ΔBE) to a precision of up to ~0.1 eV.[6] However, the definite correlation of one observed XPS peak to a given atom within a molecule and to its chemical environment is by no means straightforward. Usual procedures are based on gas phase molecular data,[7] or well-defined surface science experiments under very controlled conditions.[2,6] A more solid, unbiased identification via accurate calculations simulating the electron ionization process would be very attractive. Along this line wave function and density functional theory (DFT)-based methods have proven its suitability using the so-called ΔSCF approach,[8] in which the binding energy (BE) for a given atom CL is gained by the difference in total energy between the neutral system, and that with an a hole in the CL of the atom under inspection.

The BE values obtained through the ΔSCF approach include final state (FS) effects—that is, the excited core electron is removed from the system under scrutiny and the electronic structure modified according to the electron hole. Indeed, by considering the electronic screening of the core state and relativistic effects, one can obtain BE estimates well within the XPS experimental accuracy of 0.1 eV, either based on Hartree–Fock (HF) calculations, or using methods within DFT.[9,10] Such a high level of accuracy for isolated molecules would be desirable when they are adsorbed to a solid surface, to distinguish

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among different molecular adsorption conformations, as well as to monitor heterogeneously catalyzed processes.

Specifically, it is highly desirable to correctly model adatoms and adsorbed molecules on transition metal (TM) surfaces, given that they are ubiquitous in heterogeneous catalysis as supported active phases. One has also to keep in mind that these systems play a key role in other fields as well such as nanotechnology devices and chemical resolution components. A recent systematic study identified the Perdew–Burke–Ernzerhof (PBE)[11] and the Tao–Perdew–Staroverov–Scuseria (TPSS)[12] DFT exchange-correlation functionals as best compromises suited to simultaneously describe both main group molecules and TM surfaces.[13,14] One of the most extended ways of tackling *ab initio* studies of TM surfaces is to take advantage of periodic boundary conditions (PBCs),[15] and, especially for TM, to use pseudopotentials to describe core electrons or the projector augmented wave (PAW) method,[16] being highly accurate and in extensive use. This formalism can be regarded as an effective all electron method in which atomic cores are frozen as in the isolated atom in its spin polarized ground state. In the case one is interested in CL binding energies, the PAW description of the atomic core with a generated core hole can be used.[17] Several implementations of the PAW method exist in the literature with different specificities. Here, we focus in the implementation of Kresse of Joubert[18] in the Vienna *ab initio* simulation package (VASP) code.[19] This is a code specially designed to study periodic systems yet also suited to finite systems provided they are embedded in a periodic box sufficiently large to effectively isolate the contained species.[20]

However, periodic slab models, despite allowing for a reasonable description of surface processes on TM surfaces, hamper a rigorous application of the ΔSCF approach. This is because of two different reasons; (i) the resulting system with the core hole is periodically charged and (ii) the core hole use to be simulated with a pseudopotential rather than the full electronic description of core electrons. Despite of these difficulties, it has been shown that FS CL BEs can be estimated by generating a core excited ionic PAW in the course of the corresponding calculation, as above commented.[17] The approach neglects relaxation of core-electrons on ionization, yet valence electron screening is explicitly described. In this approach, to avoid dealing with a periodically repeated charged system, the excited electron is added to the bottom of the former conduction band—or the lowest unoccupied molecular orbital (LUMO) in the case of isolated molecules—see Figure 1. The system thus maintains its charge neutrality within the imposed PBC. While this approach seems appropriate to describe XPS with conducting probes, for isolated molecules a naïve application of the same approach would result in actually modeling a X-ray absorption near edge structure spectroscopy process. Notice that within the VASP implementation of the PAW method, absolute BEs are not accessible given that only valence energies are calculated. Other implementations exist where absolute BE can be obtained by adding up the frozen core energies,[21] although so far its use has been limited. Because of the limitation in performing proper ΔSCF in the used code, we here focus on CL binding energy shifts (ΔBEs), alleged

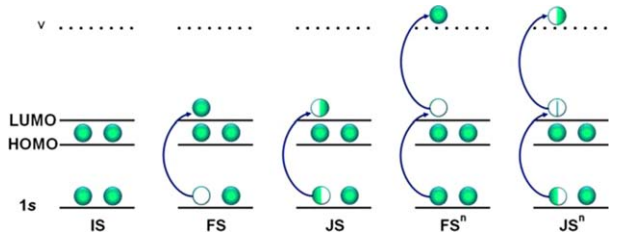



**Figure 1**. Sketch of initial state (IS), final state (FS), and Janak-Slater (JS) situations, as well as $FS^n$ and $JS^n$ approaches where excited (half) electron is removed from the system, and conceptually placed in vacuum, *v*.[Color figure can be viewed at wileyonlinelibrary.com]

however to be rather accurate[17] and widely used in practice to assign XPS signatures to certain surface species.[3–5,22–24]

The precision of this particular FS approach has been estimated to be of 0.02–0.05 eV in test calculations on selected systems.[17] Subsequent studies reported larger deviations,[3–5,24] and, in practice, different approaches are frequently considered, seeking for the best applied approach, as the accuracy issue remained unclear. Other approaches include initial state (IS) approximations, which considers that Kohn–Sham (KS) eigenstate energies in the electronic ground state as approximations to BE —similar to Koopmans theorem in HF, a statement recently discussed unambiguously showing that while Koopmans theorem does not hold for the KS eigenvalues, it properly reflects the trends in ΔBEs.[25] The proper way to define IS BEs from DFT calculations has also been described by some of us.[26] An alternative approach is provided by the model known as Janak–Slater (JS) transition state method,[27] which, in a nutshell, considers half occupation of the CL when estimating BEs rather than a full core-hole as in the FS approach, see Figure 1.[28] Here, the orbital energies are used as estimates for CL BEs and total energies are actually disregarded.

To address the accuracy of the these procedures above described, here we take advantage of the recent evaluation of PBE and TPSS performance on estimating ΔSCF BEs and ΔBEs for 1*s* CL for a large dataset of 68 molecules containing a wide variety of functional groups for first row main group elements B→F, considering up to 185 CLs, for which accurate gas phase experimental BEs are known, see Table S1 in Supporting Information.[9,29–32] Only these functionals have been assessed, as the ultimate goal is to use them to assign XPS features of atoms and/or molecules in different positions and/or conformations on TM supports. The ΔSCF calculations for this large set of molecules have been previously carried out using a full all electron description and a near HF limit basis set, obtained within spin restricted open-shell formalisms for both HF and KS methods.[9]

## Computational Details

In the present calculations dataset, the molecules were isolated in an asymmetric box with PBC ensuring a minimum vacuum distance in any direction of 10 Å. Calculations were carried out at the $\Gamma$-point and using a cut-off kinetic energy limit of 415 eV for the plane-wave basis set. The PAW approach was

used to treat core electrons and PAW core orbital energies were used to deduce the BE shifts in all cases. An ionic convergence criterion of 0.01 eV $Å^{-1}$ was used, and an electronic criterion of 10-6 eV, used together to a first-order Methfessel-Paxton smearing of 0.1 eV width. Calculations were performed in a spin-polarized and a non-spin-polarized fashion, although spin-polarization effect on BEs was found to be essentially negligible, and so for the ΔBEs.

Note that spin polarization is a necessity to properly estimate absolute BEs, either using HF or KS formalisms. Indeed, its neglecting by carrying out spin restricted HF or KS in which the core hole is evenly distributed among $\alpha$ and $\beta$ spin-orbitals, yields deviations of core orbital energies of up to ~10 eV,[21] an error remediated by carrying out calculations either spin unrestricted or open-shell. Despite of this, even if BE values obtained from ΔSCF calculations may be affected by spin polarization effects, there is evidence that the effects on the ΔBEs are negligible.[33] Consequently, only non-spin-polarized values are discussed in the following. All calculations have been carried out using the VASP program.[19]

The IS, FS, and JS BEs and the corresponding shifts, ΔBE, with respect to a given reference, see below, have been obtained following the previously prescribed methodology, but with the following modifications: (i) BEs are referred to the energy of an electron in vacuum (instead of the Fermi energy, $E_F$).[24] The Vacuum energy level is obtained in each computed case as the constant value of the electrostatic potential energy distant from the molecule. Note that this vacuum leveling has been carried out independently for each molecule, studied 1*s* level, and approximation (IS, JS, or FS based). Next, (ii) for cases involving highly-symmetry related cores different PAW types were used for these equivalent atoms to avoid delocalized electronic situations. In such cases the PAW used for the atom under evaluation was kept as the standard in use, but the PAW of the other atom(s) related by symmetry replaced by the PAW_h references.[18] Finally (iii) for FS and JS, instead of the usual addition of the excited electron —or half electron in the JS case— to the LUMO, a more physically sound situation has been contemplated, in which the excited (half) electron is removed from the system, hereafter named $FS^n$ and $JS^n$ methods, see Figure 1. The resulting charge is compensated inside the PBC box with an underlying countercharge background distributed within the box. The $JS^n$ approach thus closely corresponds to JS transition state approaches within PBCs. Furthermore, by comparison, this approach allows estimating the artificial screening effect of (half) electrons in the LUMO in the FS and JS approaches. The possible effect on ΔBEs of the box-diluted countercharge has been evaluated by doubling the cell dimensions, that is, minimizing the background countercharge density, with negligible variations—~0.03 eV.

## Results

The BEs obtained using the JS, FS, $JS^n$, and $FS^n$ protocols on PBE and TPSS density functionals are listed in Table S1 in the Supporting Information, whereas those obtained at the IS level are listed in Supporting Information Table S2. The ΔSCF approach using the PAW DFT total energy values (not shown) give very discrepant results with respect to experiments, since, as commented, only the valence electron density energy is used when core electrons are described by the PAW pseudopotentials. However, IS BE values (i.e., ground-state core orbital energies), despite being heavily and unphysically underestimated, capture the experimental BE trends (see Figure S1 in Supporting Information), as expected from previous works.[25] The shift with respect to experiment, estimated according to mean error (ME) and mean absolute error (MAE), shown in Table S3 in Supporting Information, backs up an arbitrary shifting procedure sometimes used when attempting at comparing these calculated values with experimental measurements.[17]

The underestimation of absolute binding energies by the ground-state core orbital energies in the IS approach—implying a negative relaxation energy in response to the presence of the core hole—flips to overestimation when applying any of the FS approaches—FS, JS, $FS^n$, or $JS^n$—, as exemplified for C 1*s* in Figure S2 of Supporting Information. It is worth to highlight that overall (i) PBE values are closer to experimental ones than TPSS and that ii) FS and JS values are much closer to experimental values than the $FS^n$ and $JS^n$ counterparts, yet (iii) the trend is better captured using $FS^n$ and $JS^n$ approaches, and (iv) JS and $JS^n$ values are much closer to experimental values than FS an $FS^n$. As in IS, MAE, see Supporting Information Table S4, indicates the presence of a systematic deviation which allows for shifting BE values to align themselves with the experimental ones.

The not so unexpected inaccuracy of absolute BEs, which can be, in the best cases, of up to few eV, cancels however when estimating CL binding energy shifts. To illustrate this point, ΔBEs were separately gained for main group elements B→F with respect to a given simple reference molecule: diborane ($B_2H_6$), methane ($CH_4$), ammonia ($NH_3$), water ($H_2O$), and fluoromethane ($CH_3F$) for B→F, respectively. A comparison of the performance for FS, JS, $FS^n$, and $JS^n$ in calculating ΔBEs versus experimental values is provided in Figure 2.

Note, for instance, how FS values tend to be underestimated at both the PBE and TPSS levels, with PBE values slightly closer to experimental values, although with a clear wider dispersion. When applying the JS approach, values walk the line, and so the MAE decreases, see Table 1, although the slightly better performance of PBE versus TPSS remains, although the underestimation seems to vanish. However, the MAE values for PBE and TPSS within the FS and JS approaches are 0.9 eV and 1.0 eV, respectively, and so, of little use when trying to distinguish species/conformations within the aforementioned experimental precision of 0.1 eV.

The situation is greatly improved when removing the excited (half) electron from the system, for example, when carrying out $FS^n$ and $JS^n$ approximations. The removal of the ejected electron clearly lines up ΔBEs estimates either at the full core-hole ($FS^n$) or within the JS ($JS^n$) approximation. However, deviations are still noticeable in the $FS^n$ case, especially at PBE level, with an underestimation of −1.4 eV, which drops to 0.85 eV at TPSS level. Notice how the artificial valence (half) electron screening by PAW CL estimation increased errors by 0.19

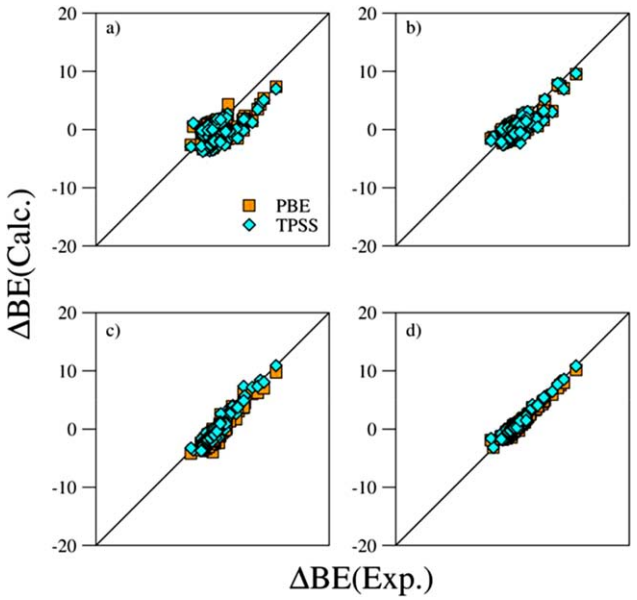



**Figure 2.** Calculated PBE and TPSS ΔBEs, ΔBE(Calc.) versus experimental values, ΔBE(Exp.), for a) FS, b) JS, c) $FS^n$, and d) $JS^n$ methods. All values are given in eV. [Color figure can be viewed at wileyonlinelibrary.com]

and 0.80 eV at PBE and TPSS levels, respectively, as observed by comparing FS and $FS^n$ values.

However, the most important result is that the half excited electron removal in $JS^n$ not only lines up estimates, but also deviations. As a result, no clear underestimation can be discerned, and, moreover, MAE values diminish to 0.37 and 0.21 eV for PBE and TPSS $JS^n$ levels. As found for $FS^n$, the half excited electron screening was responsible of an average increase of errors in ΔBEs of 0.53 and 0.8 eV for PBE and TPSS, respectively. More importantly, the ΔBEs MAE at PBE and TPSS levels of 0.37 and 0.21 eV are of the same accuracy of corresponding relativistic ΔSCF calculations of 0.24 and 0.25 eV as obtained in an earlier work.[9] Hence, for gas phase molecules, the $JS^n$ method in combination with a PAW core description is capable to reach the same high degree of accuracy for ΔBEs than all electron ΔSCF using a fully uncontracted near HF limit basis set.[9] Note that for thermochemistry of gas phase molecules hybrid functionals, such as Becke–Lee–Yang–Parr (B3LYP), are slightly better than PBE or TPSS, with a MAE for ΔBEs of 0.16 eV, as obtained from relativistic ΔSCF calculations on a subset of present data,[10] although its use is not advised for treating TM systems, as the proper metallic bond delocalization is hindered when using hybrids.[14] Last but not least, one has to keep in mind that small, yet frequent, experimental BE shifts which fall within the PBE and TPSS $JS^n$ accuracies cannot be correctly backed up by such calculations, and improved accuracy methods should be used, as the above commented. In any case, TPSS seems to be the best compromise and better suited when trying to link experimental ΔBEs to *ab initio* estimations, both reaching a similar degree of precision. Note that the superior performance of the $JS^n$ method (used with an PAW core description and PBCs here) may not be surprising based on its analogy to the original Slater transition state theory.[28]

**Table 1.** Summary of the ΔBE(Calc.) statistical analysis for all the core level studied at PBE and TPSS performance.

| | PBE | | | |
|---|---|---|---|---|
| | FS | JS | $FS^n$ | $JS^n$ |
| ME | −1.03 | −0.28 | −1.40 | −0.21 |
| MAE | 1.62 | 0.90 | 1.43 | 0.37 |
| | TPSS | | | |
| | FS | JS | $FS^n$ | $JS^n$ |
| ME | −1.21 | −0.42 | −0.85 | 0.02 |
| MAE | 1.73 | 1.01 | 0.93 | 0.21 |

All values are given in eV.

## Conclusions

In conclusion, we addressed the accuracy of full core-hole (FS) and JS approaches as implemented in VASP in predicting CL binding energies and, more specifically, CL binding energy shifts from core orbital energies using a PAW description of the atomic cores, and under PBCs, on a broad dataset of 68 molecules containing first row atoms (B→F) in a great diversity of chemical environments, accounting for a total number of 185 different 1*s* CLs. In addition to current procedures, we contemplated removing the excited (half) electron from the molecular LUMO in the here called $FS^n$ and $JS^n$ approaches. As expected, the PAW total BE estimates predicted by the VASP code do not match CL BEs as core electron energies are not explicitly accounted for. However, computed ΔBEs estimates, that is, relative BE shifts, line up well with experimental trends. For the PBE and TPSS exchange-correlation functionals, ΔBE estimates using the $JS^n$ approach yield MAEs of only 0.37 and 0.21 eV, respectively, and are thus of similar size to all electron ΔSCF calculations with saturated basis sets,[9] thereby approaching the experimental accuracy of HR-XPS of 0.1 eV. The accuracy of the $JS^n$ approach is useful not only for molecules but also promises its use in assigning ΔBEs of molecular moieties on material surfaces, with concomitant applications in surface science, nanotechnology, and heterogeneous catalysis, although its accuracy in such systems remains to be confirmed.

**Keywords:** X-ray photoemission · core level binding energy shifts · projector augmented wave · density functional theory · molecular dataset

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Additional Supporting Information may be found in the online version of this article.

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